

APPLICATION FOR SWEDISH PATENT

Title: SOUND VALVE SPEAKER FOR REGULATING PRESSURE

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SOUND VALVE SPEAKER FOR REGULATING PRESSURE

FIELD OF THE INVENTION

[0001] The present invention relates to user's need of sound, silence and air in locale with controlled ventilation or forced air circulation. It improves both the energy efficiency, frequency range, audio fidelity and usability compared to traditional loudspeaker, active noise cancellation and ventilation control systems.

BACKGROUND OF THE INVENTION

[0002] Present art ventilation control systems, particularly in modern energy-efficient buildings, are designed to be highly efficient air movers but are inherently noisy and difficult to tune to user's actual need of air. This leads to installments with either large dimension both in pipings, fans, sound blockers, sound absorbents and air diffusers, or a persistent noise that is often peaking in the 10-100 Hz range. Active noise cancellation using low-cost speakers does not cover this range. Also, the air circulation does not follow the user's need but is to a large extent wasted in empty areas. Together, this lead to unhealth and large environmental impact per person, particularly in large locale.

[0003] Present art loudspeakers are designed to be highly linear air vibrators, but have several inherent limitations:

1. Limited frequency range per vibrating membrane, leading to complex speaker constructions and arrangements that intrude on the living area.

2. Membrane breakup into multiple resonances at higher frequencies, leading to a disharmonic reproduction of sound.
3. Multiple membranes introduce phase shift, leading to unstable reproduction of sound source locale.
4. Power efficiency is ~10% in a 1-20 kHz band but falls to ~1% at 200Hz and practically zero at 20 Hz.
5. Frequencies below 50 Hz is usually only reproduced to ears in headset speakers or to the body in special chairs.

[0004] This means the user need of sound and silence is always compromised, particularly in large locale. For example, frequencies below ~100 Hz can be dropped completely which adds a sensation of “boxiness”. That frequency content can also be bundled together into a single speaker cabinet resonance adding “rumble”. It can also artificially be shifted into higher harmonics adding “tension” to the sound.

[0005] Another example is that sound reproduction in the middle to high hearing frequencies changes dramatically depending on position due to phase shift between membranes as well as room resonances.

[0006] What is needed is a better means of delivering air, sound and silence to the user, and to achieve this without cluttering the locale with bulky installments.

OBJECTS OF THE INVENTION

[0007] The object of this invention is to deliver controlled air, improved sound and complete silence efficiently and invisibly to the user.

SUMMARY OF THE INVENTION

[0008] In order to overcome the problems with bulky and noisy ventilation systems that wastes air on empty spaces, we do the following:

1. Employ noise and presence detection by measuring sound both from inside the ventilation and from the user's mobile device.
2. Feeding a combined air and noise cancellation signal to a membrane valve that we make linear from 0 to ~200 Hz frequency.
3. Cancel frequencies above ~200 Hz using the same cancellation signal, by an air vibration action from the same membrane.

[0009] This membrane valve replaces the normal vent(s) of the locale without any need of floor space. It can be a drop-in replacement for the existing vents, be invisible to the occupants, and self-powered using an impeller hidden in the valve. It uses standard wireless, audio and motion-tracking features in the occupant's existing mobile devices for its control. Fresh air and silence follows the user because noise from the ventilation system and unwanted air is cancelled out - the former at the user's location and the latter at empty areas. Since installation time can be less than a minute, it immediately leads to improved air freshness, reduced ventilation demand, as well as less noise from the ventilation system and neighbouring rooms connected to it.

[0010] In order to improve on the problems with frequency-limited, complex, disharmonic, unstable, inefficient and intruding loudspeaker systems, we use the silent vent above [0008]. We do the following:

1. Add a wanted sound signal to the noise cancelling - turning the vent valve it into a calibrated, deep-frequency and noise-free speaker.
2. Membrane breakup is pushed to higher frequency through a stabilizing air pressure, leading to a raised upper frequency limit and less disharmonics.
3. The phase shift between speakers is eliminated, leading to a stable and more correct perception of sound source locale.
4. Frequencies below ~200 Hz are generated at high sound pressure from the vented air - without consuming any extra energy.
5. The installation is simple [0009] and non-intruding.
6. Functionality for follow-me air, follow-me silence, follow-me sound and channel separation is controlled by the user just moving around naturally.

[0011] The valve membrane will vibrate air in front of it like a normal loudspeaker at typical loudspeaker frequencies, and simultaneously modulate the air flow for generation of sound at lower frequencies. That low-frequency sound pressure is only limited by the capacity of the ventilation system, which for most locale is > 120 dB. Typical installations can have a flat ~80 dB frequency response in the full sound frequency range 10 Hz to 20 kHz (including both body motion sensing and hearing) using only two motors coaxially mounted in the valve: One outer motor that drives a valve membrane from DC to ~ 2 kHz in open-baffle and one motor that drives a

front tweeter membrane between ~ 2 kHz and 20 kHz. A method for driving both these membranes using a single motor is also described but has not yet been tested.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] For a more complete understanding of the invention, reference is made to the following description and accompanying drawings, in which:

[0013] FIG. 1 Details of a speaker valve;

[0014] FIG. 2 Main parts in an entry vent speaker;

[0015] FIG. 3 Main parts in an exit vent speaker;

[0016] FIG. 4 A pressurized duct or cavity with entry and exit valve speakers;

[0017] FIG. 5 Entry vent speaker using rear-facing passive bass reflex membrane;

[0018] FIG. 6 Exit vent speaker using forward-facing passive bass reflex membrane;

[0019] FIG. 7 Entry vent speaker using speaker surround;

[0020] FIG. 8 Entry vent speaker using diffusor-shaped passive bass reflex membrane;

[0021] FIG. 9 Ventilation and sound control system;

[0022] FIG. 10 Presence-controlled system;

[0023] FIG. 11 Lift-motion controlled system;

[0024] FIG. 12 Lowering-motion controlled system;

- [0025] FIG. 13 Approaching-motion controlled system;
- [0026] FIG. 14 Leaving-motion controlled system;
- [0027] FIG. 15 Horizontal circle-motion controlled system;
- [0028] FIG. 16 Horizontal ellipse-motion controlled system;
- [0029] FIG. 17 Vertical circle-motion controlled system;
- [0030] FIG. 18 Up-motion controlled system;
- [0031] FIG. 19 Down-motion controlled system;
- [0032] FIG. 20 Ventilation and sound feedback method;
- [0033] FIG. 21 On-off control of entry and exit valves at 20 Hz for a 12/hr air exchange rate;
- [0034] FIG. 22 Sinusoidal control of entry and exit valves at 20 Hz for a 12/hr air exchange rate;
- [0035] FIG. 23 Typical total-energy-optimized cross-over filters and resulting membrane motion (excursion) for standard n and dB requirements for a living room (5/hr, 80 dB), restaurant (12/hr, 70 dB) and library (4/hr, 50 dB);
- [0036] FIG. 24 Geometry, coordinates and streamlines for a cross section of the flow between membrane and wall;
- [0037] FIG. 25 Optimized wall shape and resulting valve action; and
- [0038] FIG. 26 Result of feedback control algorithm $zp()$ on noise cancellation and sound generation.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0039] The arrangement in FIG. 1 shows the inner workings of a sound valve capable of cancelling or generating audible as well as infra-sound and static pressure. It shows a duct wall [1] with inlet [2] and outlet [3] ends, and a leight-weight membrane [4] for controlling the flow of a fluid [5] such as air. When the membrane [4] is moved closer, further from or vibrating towards the duct wall [1], there is a closing action [6], opening action [7] or vibrating action [8], respectively, of the flow from inlet [2] to outlet [3]. This valve action is made laminar and linear to the membrane [4] position, as described later. A membrane actuator [9] such as a voice-coil, magnet and spider suspension arrangement is required. It must be capable of rapid and precise positioning of the leightweight membrane [4] from DC to audible frequencies. Somewhere connected to either end of the duct wall [1], a compressor means [10] is also required for generating a generally constant absolute pressure on that end of the duct wall [1]. The resulting pressure acting on membrane [11] adds tension that pushes breakup modes to higher frequency and lower amplitude, which therefore makes the membrane [4] a better acoustic element. Air flow around the membrane [4] edge can create an air pillow [12] that counteracts side-movements and therefore make the motion more precise. When that motion is vertical, the precision is further increased due to lack of gravitational pull sideways. When the membrane [4] surface is mostly perpendicular to the transferred flow [13], air swirls [14] with nearly the same rotational air speed as the transferred flow will exist on each side of the membrane [4], and these act as local reservoirs for flow rate changes. These flow reservoirs are important to faciliate in the design since

they make the operation quieter and faster: With reservoirs rapid flow rate changes happens through small and laminar air direction changes (quietly) for good sound valve operation at high frequency. Without them the rapid flow rate changes happens by abrupt and therefore turbulating air speed changes (noisy) so that good sound valve operation is limited to lower frequencies. These generic design requirements are summarized in claim 1.

[0040] The arrangements in FIG. 2 and FIG. 3 shows exemplary arrangements of a preferred embodiment for noise-free inlet air to a room and outlet air from a room, respectively. In FIG. 2, one sees a vent tube [17] with a preferable expansion of the inner wall in the air flow direction. A speaker [16] assembly with its membrane in a preferable orientation against the air flow is held by a frame [15] so that the speaker membrane edge is centered in the tube [17] expansion. Air flow is designed to be linear with the membrane position as described later. The membrane motion can cancel incoming pressure variations and also control the static pressure and air exchange rate in the room. An optional fan [18] assembly is used for energy harvesting and pressure boosting. In FIG. 3 the frame [15] functions as decoration and dust filter for the speaker [16] assembly which in this case is held by a hidden grip inside the tube [17].

[0041] The arrangement in FIG. 4 shows similarly exemplary arrangements of preferred embodiments for stand-alone noise cancellation (claim 5) and sound generation (claim 6) using one or more pressurized duct or cavity. The stand-alone unit can be for example an air conditioner, a vacuum cleaner, an air purifier/dryer/humidifier, a wall or speaker cabinet, or a duct between two far ends of an arena. It can use any volume large enough to accomodate a substantial amount of pressurized air, or any duct with ends farther apart than the longest wavelength of sound to be con-

trolled, as described later. The suction speaker [19] controls air passing through a preferably centrifugal and/or forward-bent fan [18] (claim 7) which pressurizes a duct or cavity [20]. A pressure speaker [21] controls air release from the duct or cavity [20].

[0042] The arrangements in FIG. 5 and FIG. 6 shows similarly exemplary arrangements of preferred embodiments for full-spectrum sound generation in a room - the former for inlet air and the latter for outlet air. The frame [15] holds a speaker [16] assembly containing a front-firing speaker and a rear passive bass reflex membrane that protrudes into the tube [17] expansion. Back pressure variations in the speaker [16] box result in hydraulic motion of the protruding membrane. This low-frequency motion changes the air flow aperture and the resulting room pressure, resulting in increased low-frequency sound and noise cancellation as described later. The design requirements are summarized in claim 10.

[0043] The arrangements in FIG. 7 and FIG. 8 shows similarly alternative arrangements for sound generation from incoming air into a room - the former for lower frequency ventilation and noise control, and the latter for full-spectrum sound. In FIG. 7, the frame [15] holds a rear-facing flexible surround of a speaker [16] assembly that is mounted in an inlet tube [17]. The air valve action is here between the rear of the speaker surround and a fixed plate. In FIG. 8, the frame [15] holds a speaker [16] assembly with passive bass element that act as diffusor for incoming air. The air valve action is here between the diffusor membrane and a fixed tube [17].

[0044] The system in FIG. 9 and method in FIG. 20 illustrates auxiliary devices, signal processing and applications needed for precise control of air flow, noise cancellation, and full-spectrum sound in a room according to the present invention. In FIG. 9 which is used in claim 12, the system [22] contains one or more sound valve [23]

wirelessly connected to one or more mobile device [25]. The latter is used to control the ventilation, noise cancellation and apparent sound source [24] generation by its presence (FIG. 10 used in claim 13), motion and gestures as detailed in claim 14: Lifting (FIG. 11) and lowering (FIG. 12) enables and disables sound setting adjustments. An apparent sound source [24] is controlled by moving towards it (FIG. 13), moving from it (FIG. 14) and rotating horizontally elliptic towards it (FIG. 16) to maintain/increase, maintain/reduce and increase/reduce its apparant channel separation, respectively. A horizontal circular motion (FIG. 15) controls the overall audio volume. A vertical circular motion towards one sound valve [23] controls its audio volume (FIG. 17). A wave upwards (FIG. 18) and downwards (FIG. 19) switches on and off audio, respectively.

[0045] In FIG. 20 which is used in claim 15, the sound control method [26] uses one or more ventilation and sound generation method [27] such as the one in the in a sound valve [23] described above in concert with a ventilation and sound signal transmission method [28] in a mobile device [25]. The mobile device [25] maintains a calibration for both an air flow model as described later, a noise model (claim 18) and a sound generator model (claim 17) to regulate the air flow and sound reaching the mobile device [25]. The result is finer-tuned control of both the ventilation, noise cancellation and sound generation, as described later.

Other considerations [0046] A specific percussion sound with extra fast attack and deep bass, suitable for large arena use, can be generated very efficiently through claim 8. Here the membrane intentionally accelerates by increasing fan pressure and suction until it lands with an explosive sound or bounces with a smashing sound.

[0047] For narrow inlet vent designs such as those in high-pressure diffusors with

long throw, claim 9 describes an alternative. In FIG. 1, the two air swirls [14] needed for noise-free high-frequency operation are typically rather small and unstable, and can therefore be limited in capacity, but here that is not a limitation since both sides of the membrane take part in a large doughnut-shaped or eight-shaped swirl that is (in a state-of-the-art long-throw diffuser) laminar and stable by design.

[0048] Installation of sound- and air-managing vents into an existing ventilation system is inherently scalable since the performance depends on air exchange rates rather than room sizes, as detailed in claim 11. However, if the rich bass sound generation capability is used in one such vent, that bass sound spreads to all other vents in the same channel. Therefore silence needs to be taken care of first: A bass-generating vent may require all other vents connected to the same channel to use noise cancelling (claim 5).

Analysis of air flow, air volume and resulting pressure variations [0049]

The most extreme sound pressure in a ventilated room is achieved by two vent speaker valves that alternates between fully open and fully closed position (FIG. 21). The pressure gradient is then $\pm n^{\text{bar/hr}}$. A sinusoidal control (FIG. 22) result in an rms sound pressure

$$dB_{max} = 20 \log \left[\frac{n^{\text{bar/hr}}}{\sqrt{2} 2\pi f p_0} \right] \text{ dB}, \quad (1)$$

where $p_0 = 20 \mu\text{Pa}$ is defined as 0 dB (typical hearing limit for humans ears). At $12^{\text{bar/hr}}$ as typically required for a restaurant, a 20 Hz tone can reach almost 100 dB sound pressure. This valve sound generation capability falls off like pink noise: -3 dB/octave .

[0050] In order to keep the frequency response flat, the membrane position needs to follow a low-pass filtered derivative of the signal multiplied by f/n , combined with a high-pass filtered raw signal (the desired sound). The upper frequency of operation

$$f_{max} = \frac{n \text{ bar/hr}}{\sqrt{2} 2\pi p_0} \cdot 10^{-dB/20} \quad (2)$$

for valve-generated sound, is typically in the range 80-2000 Hz depending on sound pressure requirements and available air flow. Typical total-energy-optimized cross-over filters and resulting membrane motion (excursion) is shown in FIG. 23 for standard n and sound pressure requirements for a living room (5/hr, 80 dB), restaurant (12/hr, 70 dB) and library (4/hr, 50 dB). Note in FIG. 23 the very small and therefore power-efficient excursions at lower frequencies, when the sound pressure is generated by modulating large flows from efficient fans rather than by pushing only small volumes of air in front of the membranes.

Valve design [0051] A loudspeaker valve has an expansion in the wall where sound is generated by varying the air flow - larger flow as the speaker membrane moves downstream and smaller flow as it moves upstream. The following calculation is for designing the wall radius $R(z)$ of the expansion so that the resulting sound has a minimum of harmonics.

Parameter	Symbol	Typcal value
Pipe diameter	D_p	125 mm
Speaker membrane diameter	D_m	100 mm
Speaker membrane hole diameter	D_h	0 mm
Maximum membrane offset	x_{mech}	15 mm
Maximum linear membrane offset	x_{max}	10 mm

[0052] A valve geometry with coordinates and streamlines are sketched in FIG. 24.

Assumptions:

1. The driving pressure is constant.
2. Transferred streamlines [35] are parallel with the duct wall [1] closest to the membrane edge [36].
3. The flow is proportional to a valve constriction area A between wall and membrane which is orthogonal to the transferred streamlines.
4. The membrane accomodates air swirls [14] of non-transferred streamlines on each side, where flow-changes for sound generation can be grabbed and returned easily, i. e. by only a (small) change in direction for the affected streamlines near the edge of the membrane.

[0053] The valve constriction area A can then be approximated by a cut cone from an upstream position $z_w < z$ where the valve wall is perpendicular to a membrane edge at z . From the coordinates in the marked triangle [37] we get:

$$\frac{z - z_w}{R(z_w) - D_m/2} = \tan \theta = \frac{dR}{dz}(z_w) \quad (3)$$

$$S^2 = (z - z_w)^2 + \left[R(z_w) - \frac{D_m}{2} \right]^2 \quad (4)$$

From (3, 4) we obtain mappings from z_w to z and A :

$$z = z_w + \left[R(z_w) - \frac{D_m}{2} \right] \frac{dR}{dz}(z_w) \quad (5)$$

$$A = \pi \left[R(z_w) + \frac{D_m}{2} \right] \sqrt{\left[\left[R(z_w) - \frac{D_m}{2} \right] \frac{dR}{dz}(z_w) \right]^2 + \left[R(z_w) - \frac{D_m}{2} \right]^2} \quad (6)$$

[0054] Under the assumptions above, an optimal valve wall shape $R(z)$ and resulting valve opening area $A(z)$ between wall and membrane has the following properties:

Constraint	Description
$R(-x_{mech}) = D_m/2 + \delta$	Valve is almost closed in the upstream end.
$R(+x_{mech}) = D_p/2$	Valve is open in the downstream end.
$\frac{dR}{dz}(-x_{mech}) = 0$	Entry connects to axial flow (application-dependent).
$\frac{dR}{dz}(+x_{mech}) = 0$	Exit connects to axial flow (application-dependent).
$\frac{dA}{dz}(z) \geq 0 \forall z \in (-x_{mech}, +x_{mech})$	Valve opens in the downstream direction.
$\frac{d^2A}{dz^2}(z) = 0 \forall z \in (-x_{max}, +x_{max})$	Valve opening is linear with membrane position.

[0055] An optimized wall shape and resulting valve characteristics is plotted in FIG. 25. The net valve action is summarized by three parameters: nominal valve area $A_0 = 25 \text{ cm}^2$, $A'_0 = 2 \text{ cm}^2/\text{mm}$ and a drag coefficient $C_d \approx 0.9$. The resulting acoustic pressure can for a sufficiently small volume V containing n mole air be approximated by

$$p = R T \left[\frac{n}{V} - \frac{n_0}{V_0} \right]. \quad (7)$$

The net room volume and the change rate for the amount of gas molecules in the room via m_i inlet and m_o outlet valves is approximated below:

$$V = V_0 + \frac{\pi}{4} (D_m^2 - D_h^2) (m_i z_i + m_o z_o) \quad (8)$$

$$\dot{n} = \left[m_i (A_0 - A'_0 z_i) \sqrt{\frac{2 (\Delta p + p_i - p)}{\rho}} - m_o (A_0 + A'_0 z_o) \sqrt{\frac{2 (\Delta p - p_o + p)}{\rho}} \right] \frac{C_d n_0}{R T V_0} \quad (9)$$

[0056] The result of feedback control of noise cancellation and sound generation using Algorithm 1 below for some typical and extreme noise and sound signals are plotted in FIG. 26. In this example, room volume $V_0 = 2.4 \text{ m} \cdot 3.2 \text{ m} \cdot 5 \text{ m}$, air temperature $T_0 = 293.15 \text{ K}$, room pressure $p_0 = 1 \text{ atm}$, ventilation pressure $\Delta p = 50 \text{ Pa}$, air density $\rho_0 = 1.204 \text{ kg/m}^3$ and air molar mass $M = 28.97 \text{ g/mole}$.

[0057] It will thus be seen that the objects set forth above, among those made apparent from the preceding description, are efficiently attained and, because certain changes may be made in carrying out the above method and in the construction(s) set forth without departing from the spirit and scope of the invention, it is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

[0058] It is also to be understood that the following claims are intended to cover all of the generic and specific features of the invention herein described and all statements of the scope of the invention which, as a matter of language, might be said to fall therebetween.

What is claimed:

1. A valve speaker, comprising; a duct wall [1] having an inlet [2] and an outlet [3] end, and a membrane [4] for controlling the flow of a fluid [5] from inlet [2] to outlet [3] in conjunction, wherein said membrane having a closing action [6], wherein said membrane [4] abutting said duct wall [1]; such that said flow between said duct wall [1] and said membrane [4] is decreased; an opening action [7], wherein said duct wall [1] having an aperture between said membrane [4] and said duct wall [1]; such that said flow between said duct wall [1] and said membrane [4] is increased, and, a vibrating action [8] between said closing action [6] and said opening action [7], wherein said membrane [4] is configured for a laminar said flow that is linear to the position of said membrane [4] for generation of sound; and, an actuator [9] for moving said membrane [4] with respect to said duct wall [1]; and a compressor means [10], for generating a constant pressure in said inlet [2] or said outlet [3]; wherein said membrane [4] is enclosed by said duct wall [1] or said duct wall [1] is enclosed by said membrane [4]; and a user of average said fluid [5] flow; said user has requirements on pressure variations in said fluid [5].
2. Valve speaker according to claim 1, wherein said pressure stabilizes the shape of said membrane [4].
3. Valve speaker according to claim 1, wherein said flow creating an air pillow [12] that counteracts side-movements of said membrane [4]; preferably said motion of said membrane [4] is vertical.
4. Valve speaker according to claim 1, wherein the edge of said membrane [4] is in general perpendicular to the transferred flow [13], with substantial and laminar air swirls [14] on each side of said membrane [4].

5. Valve speaker according to claim 1, wherein said membrane is configured to move out of phase with an existing variation in said fluid flow.
6. Valve speaker according to claim 1, wherein said membrane is configured to move in phase with a desired variation in said fluid flow.
7. Valve speaker according to claim 1, wherein said membrane having said opening action towards said outlet; wherein said compressor comprises rotor blades having an increasing angle with respect to said fluid flow.
8. Valve speaker according to claim 1, wherein said compressor comprises rotor blades having a decreasing angle with respect to said fluid flow; and said membrane having said closing action towards said outlet; and said closing action is primarily flat; and said valve speaker is a percussion instrument, or said actuator is hand- or drumstick-operated, or said actuator is foot-pedal operated.
9. Valve speaker according to claim 1, wherein said fluid flow enters or exits an edge of said membrane; and said membrane constitutes a slot opening in said duct wall.
10. Valve speaker according to claim 1, wherein said membrane comprises a flexible connection to said actuator; and said membrane has a certain weight; and said connection is a compressible fluid; preferably said actuator is the main speaker element of a closed loudspeaker box and said membrane also constitutes its passive bass reflex element.
11. Valve speaker according to claim 1, where said compressor is a ventilation or air conditioning means, said fluid flow changes the air in the room 5-20 times per hour during occupancy; root mean square of said fluid flow variation during

said vibration reaches at most 71% of said fluid flow; preferably said valve is part of a flow-regulated ventilation system.

12. Valve speaker system [22], comprising; one or more sound valve [23] according to any of the preceeding claims; one or more apparent sound source [24] perceived by said user or group of said user; one or more mobile device [25] provided by said user or group of said user; wherein movement of said one or more mobile device [25] by said user or group of said user controlling a mode of one or all of said one or more sound valve [23].
13. Valve speaker system [22] according to claim 12, wherein said movement changes the distance of said user or group of said user to said one or more sound valve [23] and said mode is activating or deactivating said one or more sound valve [23].
14. Valve speaker system [22] according to claim 13, wherein said movement of a said mobile device [25] by a said user comprises:
 - a. a lift motion, wherein said mode is enabling a sound settings adjustment; or
 - b. a lowering motion, wherein said mode is disabling a sound settings adjustment, or
 - c. a motion towards a said apparent sound source [24] from said one or more sound valve [23], wherein said mode is maintaining the for said user apparant background audio volume and either maintaining or increasing the for said user apparant channel separation from said one or more sound valve [23]; or
 - d. a motion from a said apparent sound source [24] from said one or more sound valve [23], wherein said mode is maintaining the for said user apparant

background audio volume and either maintaining or decreasing the for said user apparant channel separation from said one or more sound valve [23]; or

e. a horizontal circle motion, wherein said mode is raising or lowering the audio volume of all of said one or more sound valve [23]; or

f. a horizontal ellipse motion towardsa said apparent sound source [24] from said one or more sound valve [23], wherein said mode is increasing or decreasing the for said user apparant audio channel separation for said audio source from said sound valve [23]; or

g. a vertical circle or vertical ellipse motion towards one of said one or more sound valve [23], wherein said mode is increasing or decreasing the audio volume of said one of said one or more sound valve [23]; or

h. a lift-and-lowering motion, wherein said mode is switching on the audio volume of all of said one or more sound valve [23]; or

i. a lowering-and-lift motion, wherein said mode is switching off the audio volume of all of said one or more sound valve [23].

15. A sound control method [26] for transmission of sound, comprising; a ventilation and sound generation method [27] transmitting a sound from a sound valve [23] according to any one of claims 1 to 11 to a mobile device [25] having microphone means; a ventilation and sound signal transmission method [28] transmitting a desired sound signal over at least one communication channel between said mobile device [25] and said ventilation and sound signal transmission method [28]; wherein said at least one communication channel are chosen from the group Bluetooth, WiFi or UWB;

16. Method according to claim 15, comprising; amplitude and/or delay in said sound in said microphone means in said mobile device [25] from said sound valve [23], and/or signal amplitude and/or signal delay in said at least one communication channel, is used to track movement and/or magnitude of movement and/or position of at least one said user.
17. Method according to claim 15, comprising; compensating said sound from said sound valve [23] by the spectral and/or temporal and/or air-flow dependency of said sound from said sound valve [23] in said microphone means in said mobile device [25]; preferably the frequency response of said microphone means is flat to < 0.1 dB in the range 10 Hz .. 20 kHz.
18. Method according to claim 17, comprising; counter-acting remaining unwanted sound in said microphone means in said mobile device [25] by generating out-of-phase said unwanted sound from said sound valve [23] at said microphone means; wherein said counter-acting round-trip is faster than the variations in said unwanted sound or faster than the variations in a model of said unwanted sound; preferably said round-trip is 2-50 ms.

ABSTRACT

A method and apparatus that delivers controlled air, sound and silence invisibly to the user is revealed (FIG. 1 to FIG. 8). A method of controlling air exchange rates in locale, cancelling fan and ventilation noise, and generating sound or silence from DC to 20 kHz using the apparatus and the microphone(s) of a mobile device is also revealed (FIG. 20 to FIG. 26).

The apparatus comprises existing or integrated ducts and fans, motorized air valves using acoustic membranes, hydraulic interconnects, air and spring suspensions, pressure sensors, wireless systems, and control logic.

User presence, motion and gesture control of the ventilation, silencing, audio volume, audio center direction and channel separation is also described (FIG. 9 to FIG. 19).

$\mathbf{zp}(n, dB_i, dB_o, dB_r, m_i, m_o, f) = \mathbf{t}/\text{ms}, (\mathbf{z}^i/x_{\max}, \mathbf{z}^o/x_{\max}, \mathbf{p}/p_{\max \text{ push}})$:

— Example time window:

— A few samples to show typical time-dependancy

$I = 50$ — Number of samples

$dt = 0.04/f$ — Sampling interval

$\mathbf{t} = 0, dt, \dots, (I-1)dt$ — Time samples

— Example noise to cancel:

— Pure tone at dB_i in inlet and dB_o dB in outlet

$$\mathbf{p}^i = p_0 \cdot 10^{dB_i/20} \cdot \sin(2\pi f \mathbf{t})$$

$$\mathbf{p}^o = p_0 \cdot 10^{dB_o/20} \cdot \sin(2\pi f \mathbf{t})$$

— Example sound to generate:

— Pure tone at dB_r dB in room

$$\mathbf{p} = p_0 \cdot 10^{dB_r/20} \cdot \sin(2\pi f \mathbf{t}) \text{ — Wanted sound}$$

$\mathbf{z}_0^i = \mathbf{z}_0^o = 0$ — Initial inlet and outlet membrane excursion

$\mathbf{n}_0 = \rho_0 V_0/M$ — Initial amount of air in room

$\Delta V(z_i, z_o) = V - V_0$ — Room volume change from (8)

$\dot{n}(p_i, p_o, p_r, z_i, z_o) = \dot{n}$ — Air flow into room from (9)

$\delta = 0.1 \text{ mm}$ — Calculation step for excursion derivatives

for $i=0..I-1$:

— Sound overshoot for differential changes in excursion

$$\varepsilon(\delta_i, \delta_o) = R T \left[\frac{\mathbf{n}_i + \dot{n} \left(\frac{\mathbf{p}_i^i + \mathbf{p}_{i+1}^i}{2}, \frac{\mathbf{p}_i^o + \mathbf{p}_{i+1}^o}{2}, \mathbf{z}_i^i + \frac{\delta_i}{2}, \mathbf{z}_i^o + \frac{\delta_o}{2} \right) dt}{V_0 + \Delta V(\mathbf{z}_i^i + \delta_i, \mathbf{z}_i^o + \delta_o)} - \mathbf{p}_{i+1} \right]$$

$\varepsilon_i' = \frac{\varepsilon(\delta, 0) - \varepsilon(-\delta, 0)}{2\delta}$ — Dependency on inlet membrane position

$\varepsilon_o' = \frac{\varepsilon(0, \delta) - \varepsilon(0, -\delta)}{2\delta}$ — Dependency on outlet membrane position

$\mathbf{z}_{i+1}^i = \mathbf{z}_i^i - \varepsilon(0, 0) \frac{\varepsilon_i'}{\varepsilon_i' + \varepsilon_o'}$ — New inlet membrane position

$\mathbf{z}_{i+1}^o = \mathbf{z}_i^o - \varepsilon(0, 0) \frac{\varepsilon_o'}{\varepsilon_i' + \varepsilon_o'}$ — New outlet membrane position

— New amount air in room

$$\mathbf{n}_{i+1} = \mathbf{n}_i + \dot{n} \left(\frac{\mathbf{p}_i^i + \mathbf{p}_{i+1}^i}{2}, \frac{\mathbf{p}_i^o + \mathbf{p}_{i+1}^o}{2}, \frac{\mathbf{p}_i^r + \mathbf{p}_{i+1}^r}{2}, \frac{\mathbf{z}_i^i + \mathbf{z}_{i+1}^i}{2}, \frac{\mathbf{z}_i^o + \mathbf{z}_{i+1}^o}{2} \right) dt$$

$$\mathbf{p}_{i+1} = \mathbf{p}_i + R T \left[\frac{\mathbf{n}_{i+1}}{V_0 + \Delta V(\mathbf{z}_{i+1}^i, \mathbf{z}_{i+1}^o)} - \frac{\mathbf{n}_0}{V_0} \right] \text{ — Actual sound in room}$$

Algorithm 1: Calculate the time-dependent inlet membrane excursion z_i and outlet membrane excursion z_o required for reproducing sound $p(t)$ in a room volume V_0 while cancelling noise $p_i(t)$ in its inlet and $p_o(t)$ in its outlet pipes.

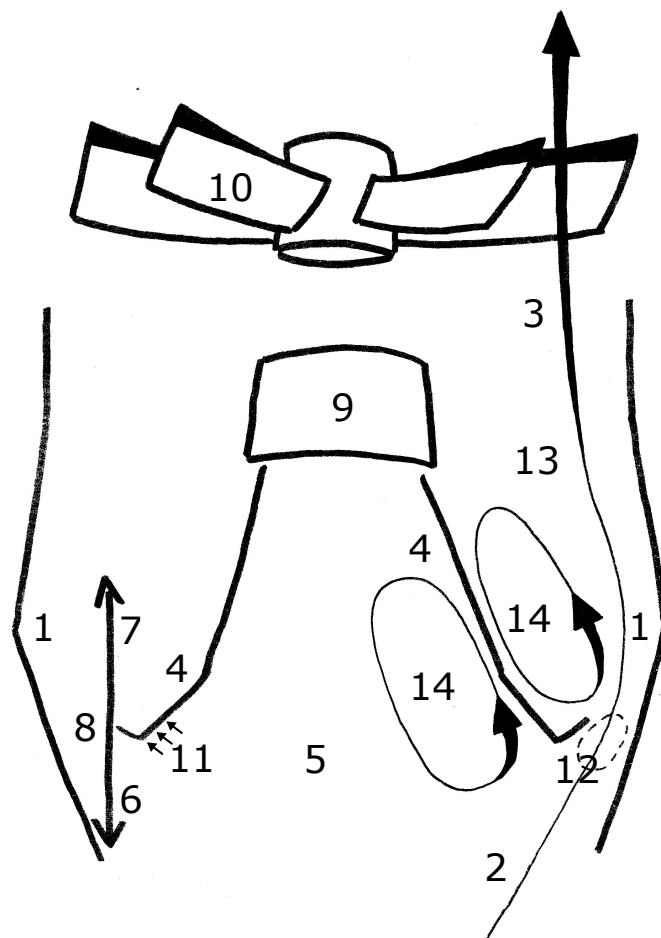


FIG. 1

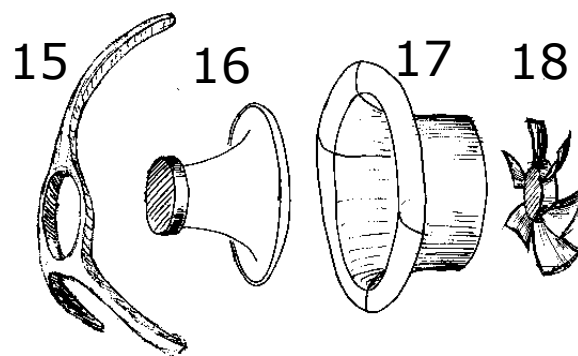


FIG. 2

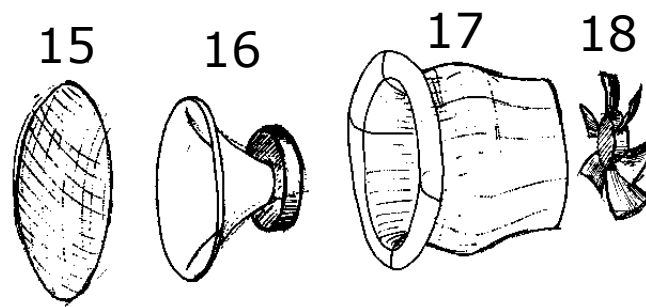


FIG. 3

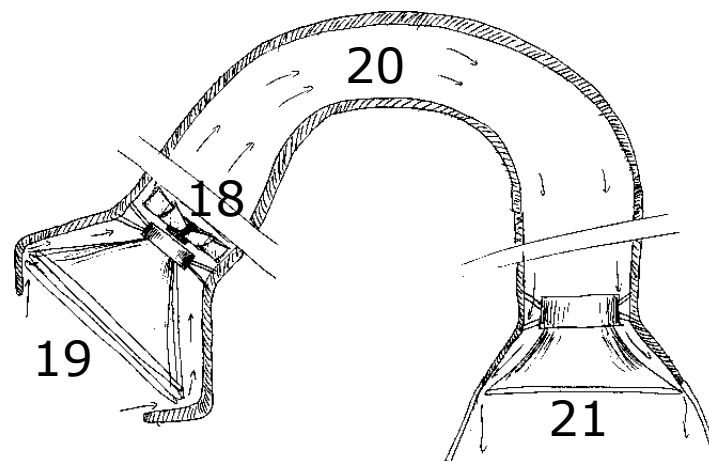


FIG. 4

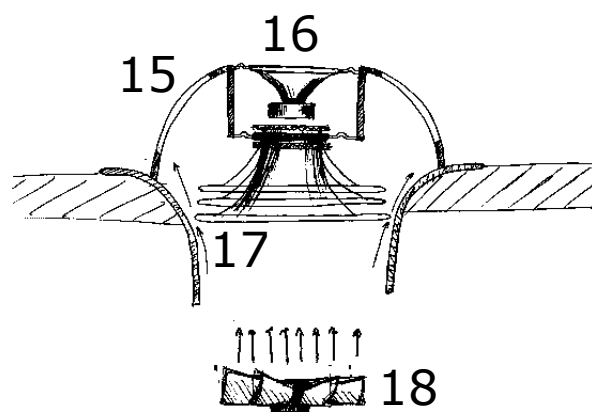


FIG. 5

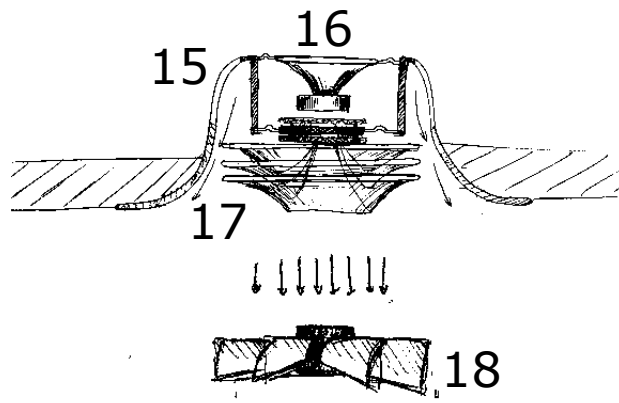


FIG. 6

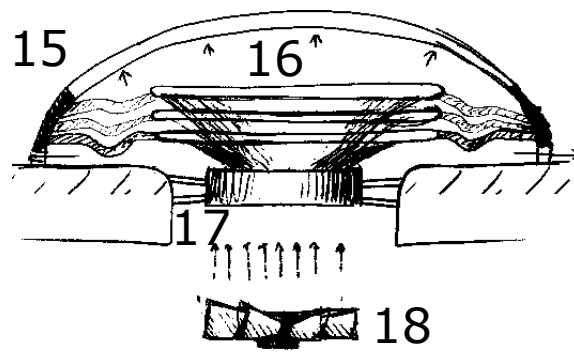


FIG. 7

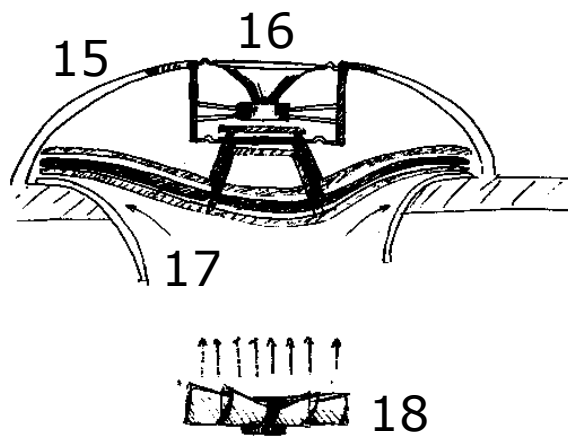


FIG. 8

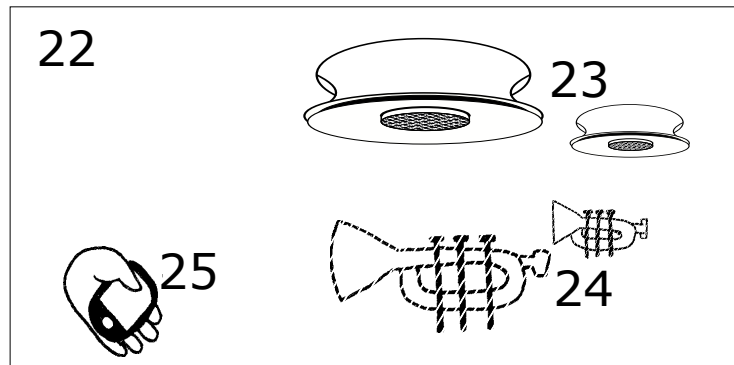


FIG. 9

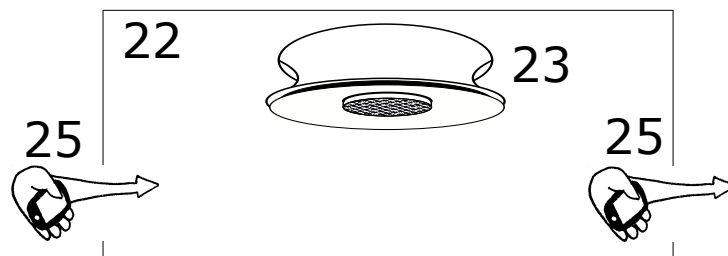


FIG. 10

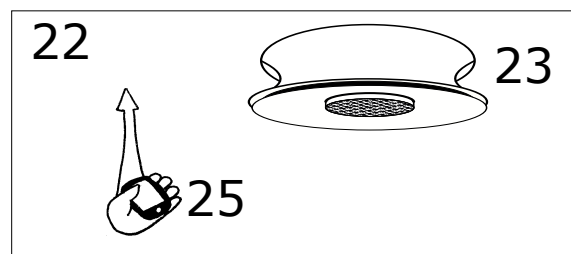


FIG. 11

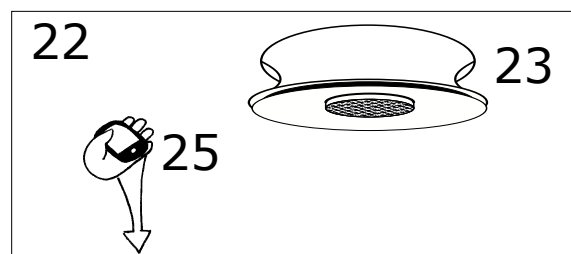


FIG. 12

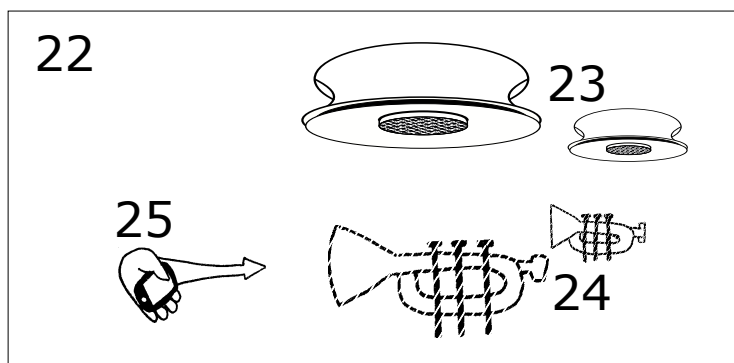


FIG. 13

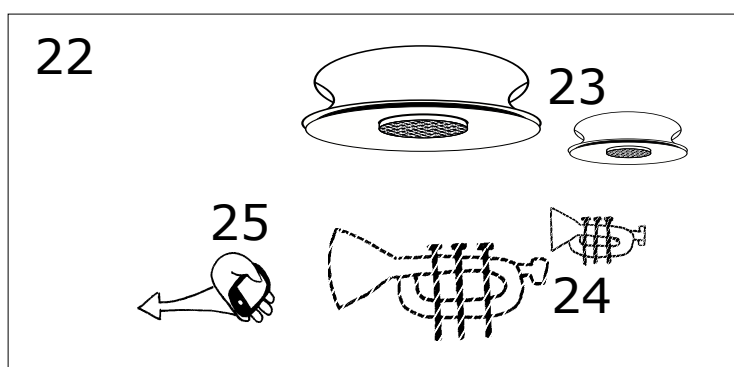


FIG. 14

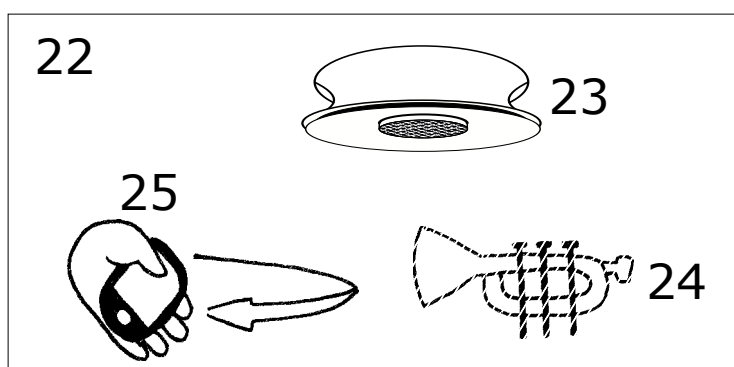


FIG. 15

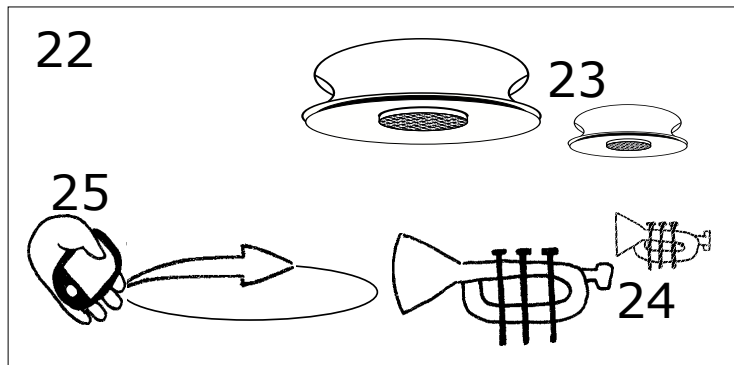


FIG. 16

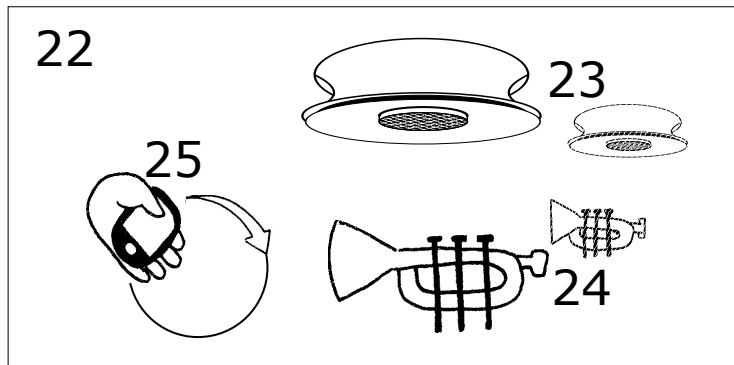


FIG. 17

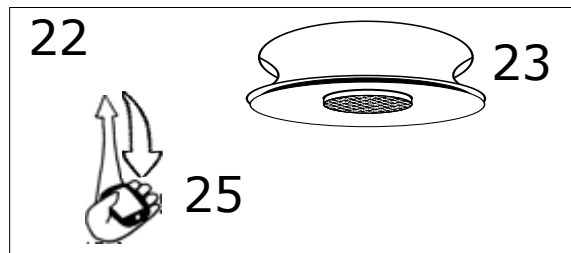


FIG. 18

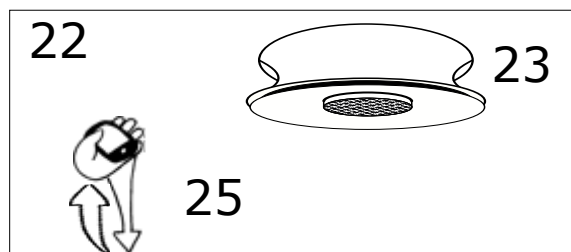


FIG. 19

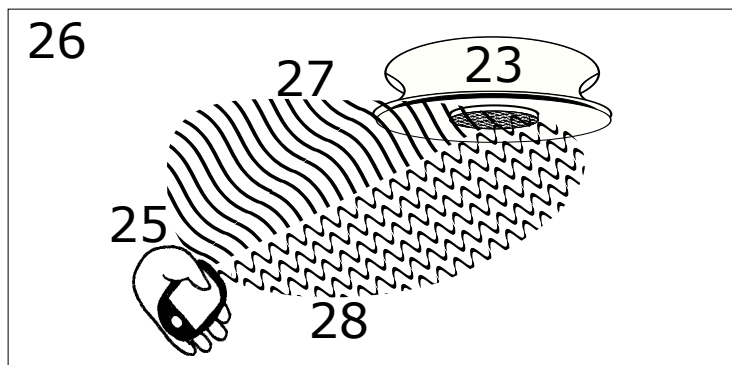


FIG. 20

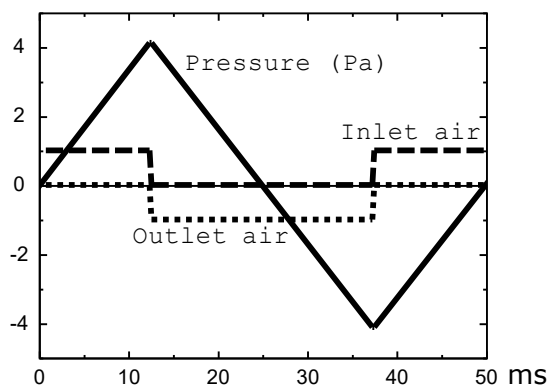


FIG. 21

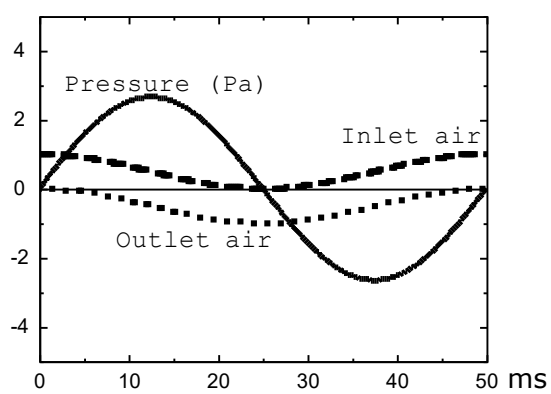
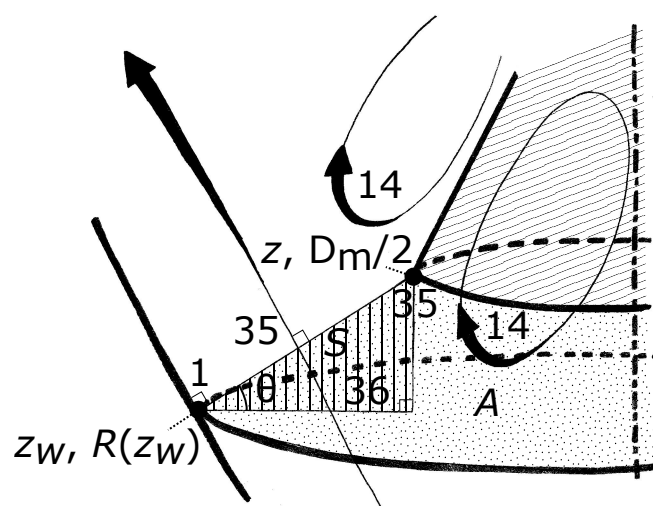
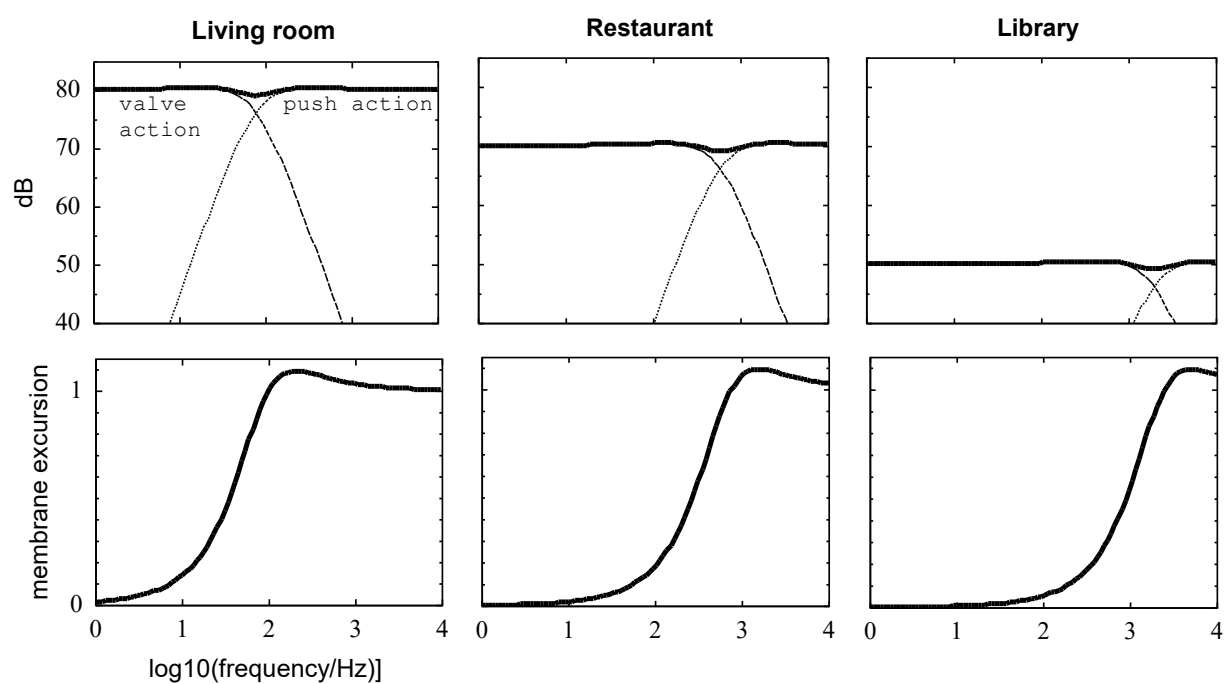


FIG. 22



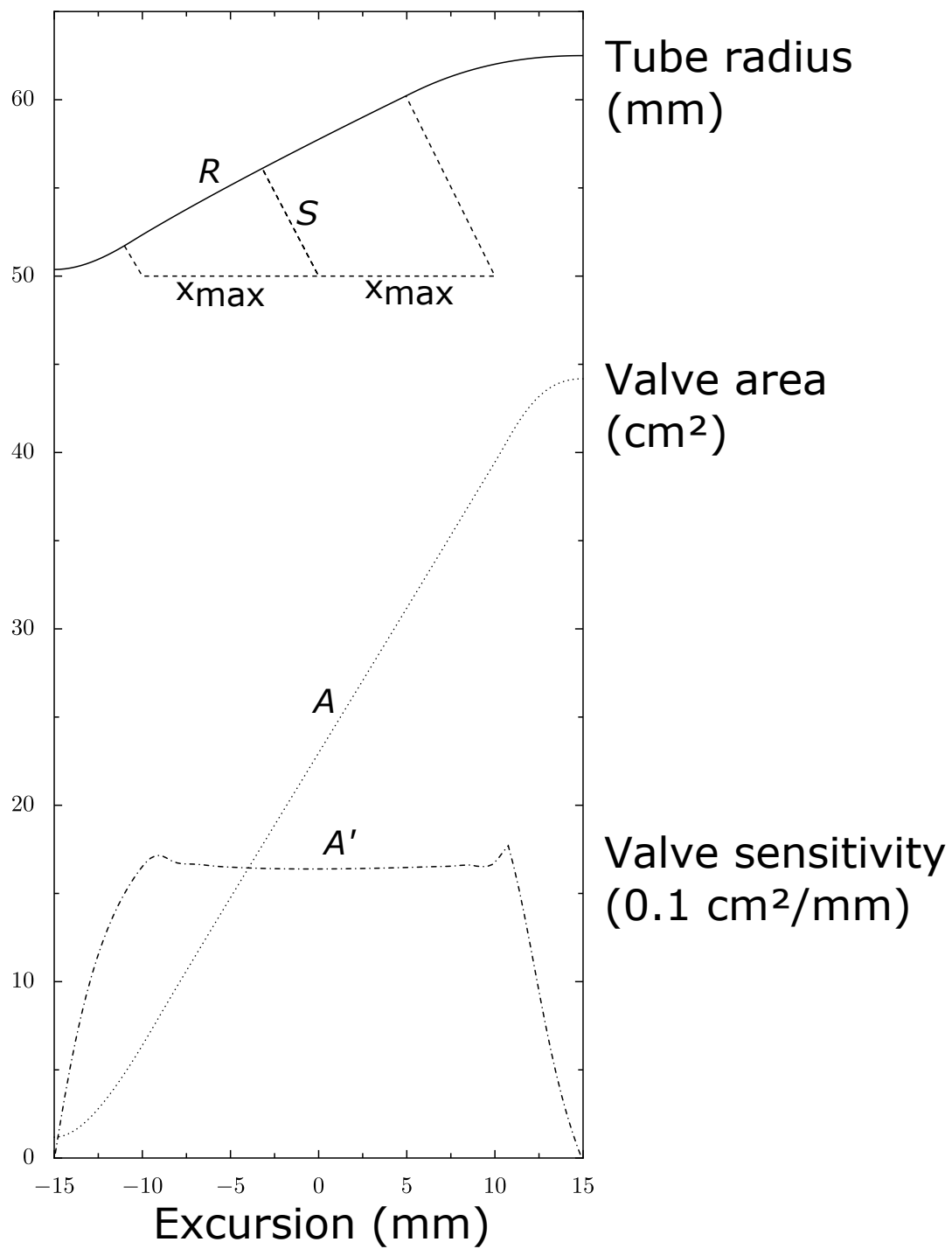


FIG. 25

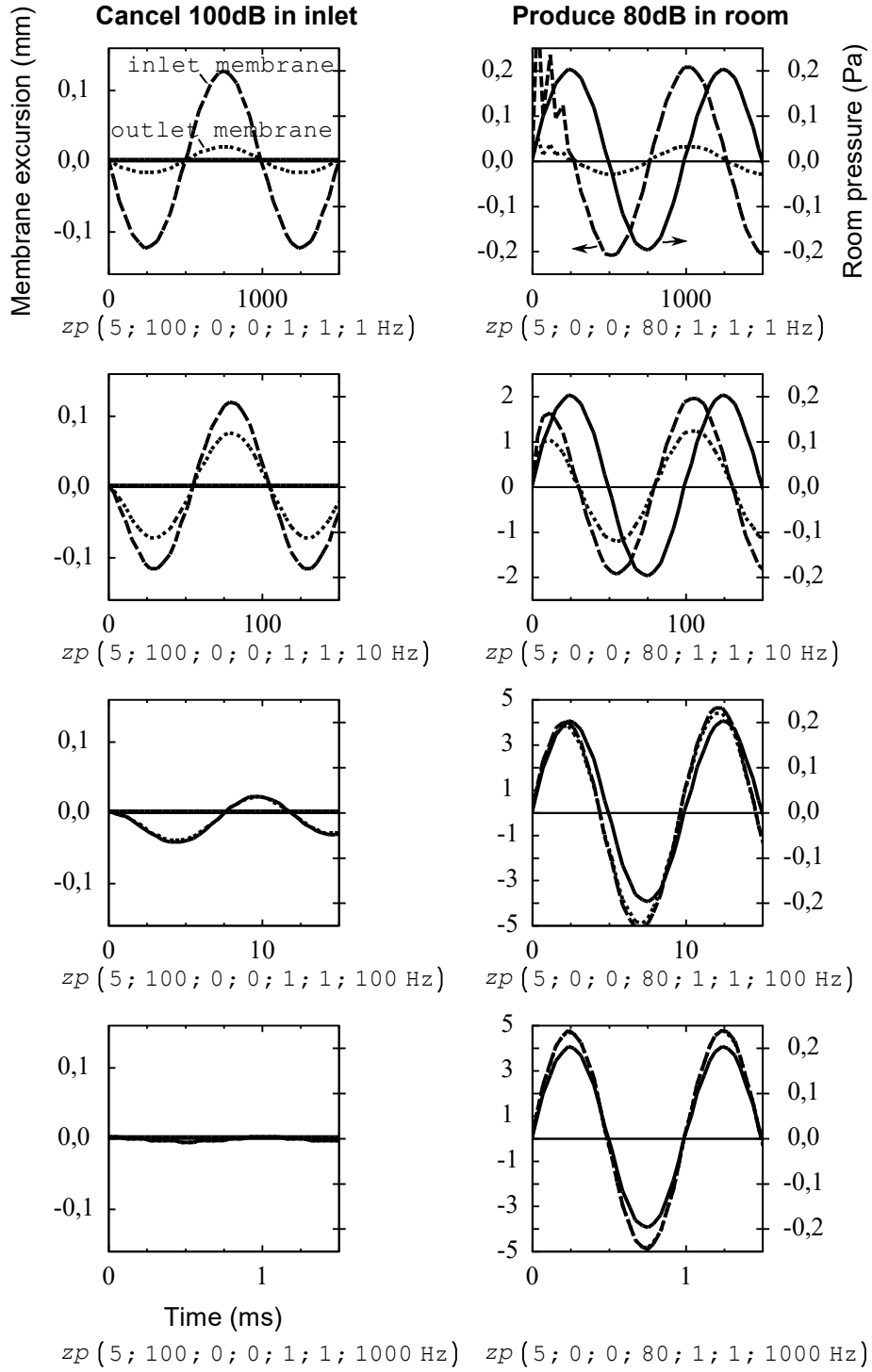


FIG. 26